

Session 5 · Deck 5.A5

Higher Order, Risk, and Pruning

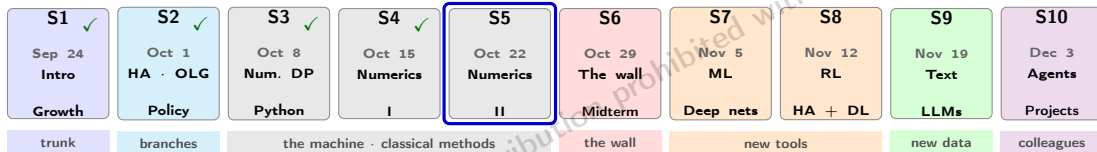
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Thursday, October 22, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

you are here: 5.A5 ▼



Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 5.A1 what perturbation computes · 5.A2 the model and the parameter · 5.A3 first order · 5.A4 linear RE systems · **5.A5** higher order, risk, pruning · 5.A6 the worked example.

Why Go Beyond First Order

The first-order solution is **certainty equivalent**:

$c_x = k_x = 0$, so the linear rules are identical in a risky economy and a riskless one. Whatever a linearized model says about uncertainty, it was assumed, not computed.

Second and higher order restore it:

- **precautionary behaviour**, through c_{xx} ;
- **risk premia** and the welfare cost of uncertainty;
- **skewness** effects, at third order.

The cost is algebra — and the method is still local, so a higher-order expansion of a badly-behaved model is a more elaborate wrong answer, not a right one.

Order buys **curvature**, not **range**. If the problem is that you are far from the steady state, going to third order is the wrong tool — that is what Block B is for.

Reference: Schmitt-Grohé & Uribe (2004) for the general second-order solution; Fernández-Villaverde, Rubio-Ramírez & Santos (2006) on what the approximation error does to likelihoods.

Second Order: Twelve Equations, Twelve Unknowns

Take second derivatives of $F(k_t, z_t; \chi)$ at $(k^*, 0; 0)$ and set each to zero:

$$F_{kk} = F_{kz} = F_{k\chi} = F_{zz} = F_{z\chi} = F_{\chi\chi} = 0.$$

Six conditions, each a 2-vector, so **12 scalar equations**.

The unknowns are the six second-order coefficients of c and the six of k — **12**, by Young's theorem, since

$$c_{kz} = c_{zk}.$$

And the system is linear. The first-order coefficients are already known and enter as constants; the new unknowns appear only linearly. Every order above the first has this property, which is why third and fourth order are tedious rather than hard.

Quant_Macro Lec. 9, "Second-order approximation"; the odd-derivative result and Young's-theorem count from FV&G, *Solution Methods V*, "Second-order approximation".

Two things vanish for free

The cross terms $c_{k\chi}$ and $c_{z\chi}$ are **zero**, and so are their k counterparts. More generally, **every odd derivative in χ is zero**. Risk enters only through even powers of the shock scale — which is to say, through variances.

The Correction for Risk

The surviving new object is

$$\frac{1}{2} c_{\chi\chi}(k^*, 0; 0) \chi^2,$$

a **constant** added to the policy function. At $\chi = 1$ it shifts consumption away from its deterministic level for no reason other than the presence of risk.

Three consequences:

- **Certainty equivalence is gone.** The policy now depends on σ .
- The **ergodic distribution** of the states moves: the economy no longer fluctuates around the deterministic steady state but around a risk-adjusted one.
- Welfare comparisons between economies with different volatility become meaningful for the first time.

A distinction worth keeping

Risk aversion is about $u'' < 0$ — the second derivative. **Precautionary saving** is about $u''' > 0$ — the third. A model can have a very risk-averse agent and no precautionary motive at all. Leland (1968), Kimball (1990); and 2.A1, where the same distinction decided who is constrained.

Measured: The Second-Order Solution

The twelve conditions of the previous frame, imposed on the quarterly model. The six numbers below are the **curvature** of the policy functions — how their slopes change as the state moves. Worst violation of any of the twelve conditions: 3.6×10^{-9} .

	$\hat{c}$	$\hat{k}'$
on $\hat{k}^2$	0.0491	0.0287
on $\hat{k}z$	-0.0890	-0.0502
on z^2	0.1082	0.0827
constant, % of s.s.	-0.00041	+0.000033

The constant is the risk correction. Consumption is **lower** than the deterministic solution says, and capital higher — precautionary saving, with a number on it.

Ledger higher_order.

Adding this entire block moves the four *first-order* coefficients by 1.5×10^{-5} . Certainty equivalence is not an approximation here; the linear terms really are unaffected.

-0.00041% is tiny because $\sigma = 0.007$ is tiny. The magnitude is a fact about the calibration; the *sign* and the mechanism are facts about the model.

Measured: The Correction Scales Like σ^2 — Until It Doesn't

If risk enters only through even powers of χ , the leading term should be quadratic in the shock size. Re-solving at multiples of σ , against the prediction “baseline $\times$ (multiple)²”:

σ	measured	$-0.00041 \times m^2$
1×	-0.00041%	—
5×	-0.00998%	-0.01022%
10×	-0.03698%	-0.04088%
20×	-0.10107%	-0.16351%

Both columns are % of steady-state consumption. At five times the shock the quadratic law is good to 2%; at twenty times it over-predicts by 60%.

Ledger higher_order.risk_sweep; all four rows solved to residual $\leq 3.6 \times 10^{-9}$.

Read the failure, not the success

The σ^2 law is a statement about the *leading* term. When the shock is large enough, the fourth-order term the expansion dropped is no longer negligible — and the second-order solution is the thing being wrong.

Extrapolating a perturbation result to a larger shock is exactly the move the method does not license.

Why Second-Order Simulations Explode

The second-order policy is

$$\hat{k}_{t+1} = b_0 + b_1 \hat{k}_t + b_2 z_t + \frac{1}{2} (b_3 \hat{k}_t^2 + 2b_4 \hat{k}_t z_t + b_5 z_t^2).$$

Simulate it by feeding $\hat{k}_{t+1}$ back in as $\hat{k}_t$. That squares a quantity that already contains a squared term, so the recursion generates terms of order 4, 8, 16, ... in the shock — **orders the approximation never claimed to be valid at.**

Sooner or later a sequence of draws pushes the state where the spurious high-order terms dominate, and the path runs away. The explosion is an artifact of the simulation, not a property of the model.

Not a stability problem

$|b_1| < 1$: the linear part is stable. It is the *quadratic feedback* that diverges, so no amount of checking eigenvalues warns you.

It also contaminates moments long before it explodes: a simulation that has not yet diverged can still have a badly wrong variance.

Pruning

The fix of **Kim, Kim, Schaumburg & Sims (2008)**: never let the simulation carry a term above the order you solved for.

Keep *two* state variables. A first-order state,

$$\hat{k}_{t+1}^{(1)} = b_1 \hat{k}_t^{(1)} + b_2 z_t,$$

and a second-order correction that is fed **only first-order terms**:

$$\hat{k}_{t+1}^{(2)} = b_0 + b_1 \hat{k}_t^{(2)} + \frac{1}{2} (b_3 (\hat{k}_t^{(1)})^2 + 2b_4 \hat{k}_t^{(1)} z_t + b_5 z_t^2).$$

Report $\hat{k}_t = \hat{k}_t^{(1)} + \hat{k}_t^{(2)}$. Nothing is ever squared twice, so no spurious order is ever created.

What it is really doing

Making the simulation consistent with the **order of the approximation**. The unpruned recursion silently claims accuracy the solution never had.

Andreasen, Fernández-Villaverde & Rubio-Ramírez (2018) give the general pruned state-space form at any order, with closed-form moments and a likelihood.

Measured: When Pruning Actually Matters

20,000 periods, same shocks, pruned and unpruned:

σ	unpruned $\max \hat{k} $	pruned
1×	0.1013	0.1013
10×	0.9947	0.9962
30×	exploded , $t = 3,967$	3.25
60×	exploded , $t = 166$	8.24

At this model's own calibration, pruning changes nothing — four significant figures agree. You need thirty times the shock before the unpruned recursion blows up at all.

Ledger pruning; second-order solves at every σ to residual $\leq 7.1 \times 10^{-9}$.

So prune anyway. It costs one extra state variable, it never hurts, and the failure it prevents is **silent until it is catastrophic** — and the threshold depends on the model, not on a rule of thumb you can carry between papers.

"It ran without exploding" is not evidence the moments are right.

How Far Away Is Any of This Valid?

Perturbation is valid within the power series' **radius of convergence**. There is a clean answer to how big that is, and it is not a real-analysis answer.

From complex analysis

The radius of convergence equals the distance from the expansion point to the **nearest singularity of the decision rule in the complex plane** — a pole, a branch point, a discontinuity in the function or any of its derivatives. If there is none, the radius is infinite.

So the trustworthy range is set by a feature of the policy function that may sit nowhere near the real line, and that no amount of staring at a simulation will reveal.

FV&G, *Solution Methods V*, "Local properties of the solution I-III".

Which is why the honest procedure is the one 3.A1 established: **compute Euler residuals off the expansion point** and look. 5.A6 does exactly that.

And the usual killer is a **boundary**: a borrowing limit or an irreversibility constraint is a non-removable singularity sitting right where the interesting behaviour is.

Regular and Singular Perturbations

There is a prior question, and it is not a technicality.

Regular perturbation: a small change in the problem produces a small change in the solution. Everything above assumes this.

Singular perturbation: a small change in the problem produces a **large** change in the solution. Then no expansion in the perturbation parameter can work, because the object being expanded is not continuous in it at zero.

Most problems in economics are regular. The standard counterexample is squarely in this course's territory: **introducing a new asset into an incomplete-market model**. Adding an asset with infinitesimal supply can change the equilibrium allocation discontinuously.

FV&G, *Solution Methods V*, "Regular versus singular perturbations".

Why this matters here

2.A1's whole point was that market structure is an *assumption*. This says the assumption cannot always be perturbed away — so "nearly complete markets" is not always near complete markets.

Nothing warns you. A singular perturbation produces numbers, and they look like numbers.

From a Point to a Distribution

Everything so far expands around a steady-state **point**:
($k^*, c^*, 0$), one number per variable.

In a heterogeneous-agent economy the aggregate state includes the **cross-sectional distribution** λ over wealth — 2.A3's object. The expansion point is then a steady-state *distribution* $\bar{\lambda}$:

$$\lambda_t \approx \bar{\lambda} + (\text{first-order deviation}).$$

The logic does not change at all. Taylor expand around a reference point; the reference point is simply no longer a scalar.

Quant_Macro Lec. 9, "From Point to Distribution: The Conceptual Extension".

Rep. agent $\rightarrow$ Het. agent	
State	$k_t \rightarrow \lambda_t$ (N bin masses)
Steady state	$k^* \rightarrow \bar{\lambda}$ (N numbers)
Expansion	$\hat{k}_t \rightarrow \hat{\lambda}_t$

Notation: the literature often writes μ for the distribution. This course writes λ for the invariant distribution and Q for the operator that moves it (2.A3).

Perturbing Around $\bar{\lambda}$: Four Steps

1. **Compute the stationary distribution** $\bar{\lambda} = Q\bar{\lambda}$ — exactly the fixed point of 2.A3, solved by iterating the Kolmogorov forward equation.
2. **Discretize it**: a histogram with N bins, so the state becomes the finite vector $(\lambda^1, \dots, \lambda^N, z)$.
3. **Linearize** the law of motion for λ_t , and the household policies around steady-state prices.
4. **Stack and apply Blanchard–Kahn**, exactly as in 5.A4. The output is first-order impulse responses for a heterogeneous-agent model.

The trade-off is a state vector of hundreds or thousands of bins — which modern linear algebra handles, because the problem is still *linear*.

Why it works

If $F(\lambda_t, z_t) = 0$ holds as an identity, we can differentiate it and match coefficients. The distribution simply has more components than a capital stock.

Reiter (2009); Auclert, Bardóczy, Rognlie & Straub (2021) for the sequence-space Jacobian, which is the same idea organised around impulse responses instead of a state vector.

Takeaways

- Second order is **12 equations in 12 unknowns, linear** given the first-order coefficients; every order above the first is bookkeeping. And every **odd** derivative in χ vanishes, so risk enters only through variances.
- The surviving object is the **risk correction** $\frac{1}{2}c_{\chi\chi}$: a constant shift that kills certainty equivalence and moves the ergodic distribution. Measured at -0.00041% of steady-state consumption, scaling like σ^2 to within 2% at five times the shock — and not at twenty.
- Second-order **simulations** feed squares back into squares and manufacture orders the solution never had. **Prune** (Kim–Kim–Schaumburg–Sims 2008).
- Measured: at this calibration pruning changes **nothing**, and at $30\times$ the shock the unpruned path explodes at $t = 3,967$. Prune anyway — the failure is silent until it is not.
- Validity is set by the **nearest singularity in the complex plane**. You cannot see it, so measure residuals off the expansion point instead.
- The expansion point can be a **distribution** — Reiter (2009), Auclert et al. (2021).

Bridge: Now Put a Number on It

Block A has claimed a great deal: fast, accurate near the steady state, useless for risk at first order, valid over an unknown range. Every one of those is a measurable statement about the quarterly model we already solved globally in 4.B5.

The next deck measures all of them — and finds the point where perturbation stops being good enough, in capital units.

Next (5.A6)

The worked example: the same model as 3.A4 and 4.B5, solved by perturbation in five steps, graded against the Session 4 global solution near and far from k^* .

And the question Block B answers: what do you do when “far from the steady state” is where the economics is?